# PROJECT TITLE: GRAMSETU (ಗ್ರಾಮ ಸೇತು)

**The Unified Rural Civic-Tech Gateway & Agronomic Intelligence Platform**

**File Naming:** GramSetu – Team GramSetu | **Team Name:** Team GramSetu (Lead: Eshwar H S) | **Category:** IT Domain

## 1. Problem Statement

Over 65% of India's population resides in rural areas, yet citizens encounter pervasive structural exclusion from modern governance portals and agricultural advisory tools. In Karnataka, over 6 crore rural citizens face acute administrative bottlenecks:

### 1.1 Rural Administrative & Linguistic Divide

Existing e-governance systems mandate high digital literacy, complex multi-step keyboard typing, and English fluency. Kannada-speaking farmers and rural elderly are disenfranchised by intricate authentication steps and lack of native speech guidance.

### 1.2 Agronomic Information Asymmetry & Crop Vulnerability

Smallholder farmers suffer devastating annual yield losses (20% to 35%) due to delayed diagnosis of crop pathogens and uncalibrated chemical pesticide usage. Extension officers cannot feasibly visit remote villages on demand, leaving farmers dependent on commercial retail distributors.

### 1.3 Escalation Vacuum In Village Grievance Redressal

Critical infrastructure failures (irrigation canal breaches, water pipeline leaks, BESCOM transformer outages) languish indefinitely due to paper records, lack of geographic isolation, and the absence of enforceable Service Level Agreement (SLA) escalation hierarchies.

## 2. Measurable Project Objectives

GramSetu establishes quantifiable, benchmarked technological goals designed to eliminate the rural digital divide:

- **2.1 Sub-Second Multilingual Voice Interaction:** Deliver speech-to-speech interaction in Kannada (kn-IN), English (en-IN), and Hindi (hi-IN) with audio synthesis latency under 800ms and conversational intent accuracy > 92%.
- **2.2 High-Precision Real-Time Crop Diagnostics:** Achieve computer vision pathology diagnosis for 15+ staple Karnataka crops (Paddy, Ragi, Arecanut, Tomato, Cotton, Sugarcane) with > 90% diagnostic accuracy and inference under 2.5s via Google Gemini Multimodal Vision with offline feature classifier fallback.
- **2.3 Automated 4-Tier SLA Grievance Resolution:** Programmatically enforce strict escalation timelines (Gram Panchayat PDO: 7 days → Taluk Executive Officer: 14 days → Zilla Panchayat CEO: 21 days → State RDPR Commissioner) to reduce average resolution time by 60%.
- **2.4 Real-Time APMC Mandi Market Transparency:** Aggregate and simulate live daily APMC price quotations across all 31 Karnataka districts with intraday volatility indicators to protect farmers from middleman exploitation.

## 3. Methodology

### 3.1 Working Principle & Architectural Logic

GramSetu is engineered as a reactive, voice-first single-page progressive web application. User vocal commands are captured via the browser Web Audio & Speech APIs, streamed into an intelligent intent router, and dispatched to either deterministic client controllers or Google Gemini generative models.

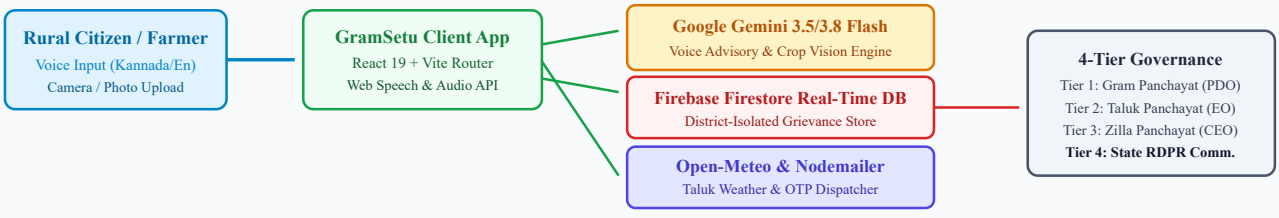



*Figure 1: GramSetu End-to-End System Architecture, AI Pipelines & Hierarchical Governance Flow*

### 3.2 AI Model Pipelines & Offline Fallback Engineering

For crop disease scanning, camera captures are compressed into base64 frames and evaluated against Google Gemini 3.5 Flash Vision. In disconnected environments or API rate-limits, a client-side Vision Feature Classifier automatically steps in, performing local pixel color distribution analysis (chlorosis, necrosis, mildew) to deliver instant agricultural triage.

## 4. Scope And Limitations

### 4.1 Operational Scope

GramSetu covers Karnataka's 31 districts, 240+ taluks, and 6,000+ Gram Panchayats. The crop database is fine-tuned for Karnataka staples (Paddy, Ragi, Arecanut, Maize, Cotton, Tomato, Sugarcane) across all four agro-climatic zones.

### 4.2 Technical Constraints & Mitigations

- **Low Bandwidth:** Handled via client-side Canvas JPEG compression (< 200KB) and localized JSON offline datasets.
- **Acoustic Ambient Noise:** Outdoor field noise during speech recognition is counteracted by acoustic transcript normalization and Kannada text transliteration.
- **Hardware Isolation:** Runs entirely on standard mobile and desktop browsers with zero requirement for specialized hardware.

## 5. Project Features

GramSetu consolidates four core functional pillars into a unified, responsive civic portal:

#### 5.1 Crop Doctor — AI Pathology Scanner

Instant diagnosis via Gemini Vision. Generates actionable reports: severity ratings, chemical dosages (Hexaconazole, Mancozeb), organic recipes, and direct PMFBY compensation links.

#### 5.2 GramSetu AI Voice Assistant

Multilingual speech-to-speech assistant supporting Kannada, English, and Hindi. Provides hands-free portal navigation and answers MSP, weather, and fertilizer questions with live waveforms.

#### 5.3 4-Tier Automated Escalation Ledger

Location-scoped grievance submission (District → Taluk → GP). Features photo evidence, automated SLA clock, tamper-proof tracking, and official administrative consoles.

#### 5.4 Live APMC Mandi Intelligence

Live district-wise APMC pricing for Arecanut, Copra, Ragi, Paddy, and Turmeric. Simulates intraday tick movements with visual bullish/bearish indicators and 5-day trends.

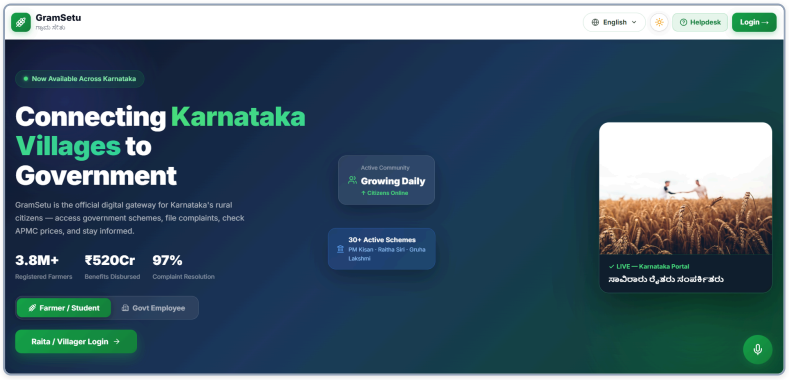



***Figure 2: GramSetu Production Working Model***
*Demonstrating the responsive Villager Portal with live APMC commodity tickers, voice assistant interface, localized Kannada UI, and instant grievance status tracking.*

## 6. Tech Stack

| Layer / Component | Technologies & Frameworks Utilized | Operational Purpose |
|---|---|---|
| **Frontend Architecture** | React 19, Vite 8, React Router v7, Vanilla CSS, Lucide React | High-performance, ultra-responsive SPAs with zero bundle bloat |
| **Artificial Intelligence** | Google Gemini 3.5/3.8 Flash, Gemini Vision, Web Speech API | Speech-to-Speech Q&A, Multimodal Plant Pathology & Translation |
| **Backend & Database** | Firebase Firestore (Realtime NoSQL), Firebase Auth | District-isolated document storage & live synchronization |
| **APIs & Cloud Services** | Open-Meteo Weather API, Nodemailer Gmail SMTP, Vercel Serverless | Real-time taluk forecasts, secure OTP emails, instant global deployment |
| **Reporting & Visuals** | Leaflet Maps, jsPDF, Puppeteer, Web Audio API | Interactive GIS heatmaps, official PDF exports, dynamic audio waves |

## 7. Innovativeness

GramSetu introduces key architectural paradigm shifts that distinguish it from legacy e-governance systems:

- **7.1 Multimodal Voice-First Rural Interaction:** Eliminates keyboard literacy requirements through end-to-end hands-free voice navigation and natural conversational Kannada speech synthesis.
- **7.2 Hybrid Edge-Cloud AI Resilience:** Combines Google Gemini Vision with an in-browser heuristic pixel classifier, guaranteeing that farmers in remote fields receive immediate crop triage even without network connectivity.
- **7.3 Automated Algorithmic Accountability:** Replaces discretionary bureaucratic handling with an automated 4-tier countdown escalation system, stripping local officials of the ability to bury citizen complaints without higher-tier oversight.
- **7.4 Hyper-Local Geographic Scoping:** Automatically filters data to "My Taluk" and "My Gram Panchayat", eliminating statewide clutter for rural citizens while isolating administrative domains for officials.

## 8. Market Potential

### 8.1 Target Audience & Commercial Viability

The primary audience comprises Karnataka's 6.1+ crore rural citizens, 78+ lakh agricultural households, and 6,000+ Gram Panchayat administrative bodies. The commercial and institutional viability includes:

- **Government Civic-Tech Procurement:** Direct institutional adoption by the Department of Rural Development & Panchayat Raj (RDPR) and Department of Agriculture for digitized grievance redressal and crop extension.
- **Farmer Producer Organizations (FPOs):** Subscription deployment for FPOs to broadcast market prices, manage collective grievances, and diagnose member crops.
- **Agri-Input & Insurance Integration:** Partnership channels with crop insurance providers (PMFBY) and certified organic bio-fertilizer cooperatives.

## 9. Scalability

### 9.1 Architectural & National Expansion

GramSetu is built on horizontally scalable serverless primitives (Vercel Edge + Firebase distributed NoSQL). The schema utilizes district-sharded collections (e.g. `district_{districtName}_complaints`), ensuring database index contention remains $O(1)$ regardless of concurrent statewide traffic surges.

### 9.2 Pan-India Multilingual Localization

The architecture decouples linguistic string tables into atomic dictionary JSON stores. Scaling to other Indian agricultural states (Maharashtra, Andhra Pradesh, Punjab) requires solely adding local language lexicons (Marathi, Telugu, Punjabi) and regional APMC mandi mappings.

## 10. Other Project Details

The complete, functional production system is deployed live on cloud infrastructure and fully open-source for academic and jury evaluation:

**10.1 Project Hosting Link (Live Working Application):**
• Live Web URL: **https://gram-setu-one.vercel.app**

**10.2 Project Hosted Source Code Link (GitHub Repository):**
• GitHub Repository: **https://github.com/eshwarhs170-a11y/GramSetu**

**10.3 Reference Links & APIs:**
• Google Gemini Developer API & Documentation: **https://ai.google.dev**
• Google DeepMind Generative Language Platform: **https://deepmind.google/technologies/gemini/**
• Firebase Realtime Cloud Firestore: **https://firebase.google.com**
• Open-Meteo Global Numerical Weather Forecast API: **https://open-meteo.com**
• Karnataka RDPR Official Administrative Portal: **https://rdpr.karnataka.gov.in**
• PM Kisan Samman Nidhi & PMFBY National Portals: **https://pmkisan.gov.in**